

SPN 521080/FMI 4 – SSAM02T

Low Beam, Left – Voltage Below Normal

Table 1, SPN 521080/FMI 4

Description	Voltage is below normal, or shorted to a low source.
Monitored Parameter	N/A
Enable Conditions	The SSAM detects voltage below normal on connector 6, pin 24.
Monitored Sequence	Continuous
Execution Frequency	Always
Typical Duration	2 seconds
HMI Indication	None
Symptom	The left low beam is inoperative.
Verification	The left low beam is operating normally.

Table 1, SPN 521080/FMI 4

Table 2, Wiring

Description	ECU	Connector/Pin	Component	Connector/Pin	Module	Circuit
LOW BEAM, LEFT	SSAM02T	6/24	LED HEADLAMP, LEFT	LOW BEAM, LEFT/1	–	–
HIGH BEAM, LEFT	SSAM02T	2/13	LED HEADLAMP, LEFT	HIGH BEAM, LEFT/2	–	–
HEADLAMP, LEFT	SSAM02T	11/1	LED HEADLAMP, LEFT	HEADLAMP, LEFT GND/3	–	–
LOW BEAM, RIGHT	SSAM02T	8/23	LED HEADLAMP, RIGHT	LOW BEAM, RIGHT/1	–	–
HIGH BEAM, RIGHT	SSAM02T	4/16	LED HEADLAMP, RIGHT	HIGH BEAM, RIGHT/2	–	–

HEADLAMP, RIGHT	SSAM02T	10/1	LED HEADLAMP, RIGHT	HEADLAMP, RIGHT GND/3	–	–
POWER SUPPLY	LED HEADLAMP, LEFT	POWER SUPPLY/8	VPDM	J7/E8	–	–
POWER SUPPLY	LED HEADLAMP, RIGHT	POWER SUPPLY/8	VPDM	J7/E5	–	–

Table 2, Wiring

Fault Code Troubleshooting

1.

☐ Turn the key ON, and connect DiagnosticLink®. Are there faults being reported by other modules?

a.

☐ **YES** → Repair as needed, then verify the repair.

b.

☐ **NO** → Go to step 2.
2.

☐ Verify that the vehicle is correctly parameterized. Is the vehicle correctly parameterized?

a.

☐ **YES** → Go to step 3.

b.

☐ **NO** → Correct vehicle parameters, then verify the repair.
3.

☐ Turn the rotary switch to the headlamp ON position. Does the headlamp turn on?

a.

☐ **YES** → Clear the fault, then verify the repair.

b.

☐ **NO** → Go to step 4.
4.

☐ Disconnect DiagnosticLink from the vehicle, and turn the key OFF. If equipped, turn the battery load disconnect switch OFF.
5.

☐ Disconnect the left headlamp assembly connector. Inspect the connector for bent, spread or corroded pins. Inspect the connector for bent, spread or corroded pins. Is there any pin damage?

a.

☐ **YES** → Repair as needed, then verify the repair.

b.

☐ **NO** → Go to step 6.
6.

☐ Check the voltage between pin 1 (LED) or pin 1 (HAL - low beam) and battery positive (+). Is voltage present?

a.

☐ **YES** → Go to step 7.

b.

☐ **NO** → For halogen headlamps, replace the bulb if needed. For Light-Emitting Diode (LED) headlamps, replace the ULB (universal LED ballast). Verify the repair.

7. ☐ Disconnect Single Signal and Actuation Module (SSAM) connector 6. Inspect the connector for bent, spread or corroded pins. Is there any pin damage?

a. ☐ **YES** → Repair as needed, then verify the repair.

b. ☐ **NO** → Go to step 8.
8. ☐ Check the voltage between SSAM connector 6, pin 24 (harness side) and battery positive (+). Is voltage present?

a. ☐ **YES** → Repair short to ground in the wiring harness. Verify the repair.

b. ☐ **NO** → Go to step 9.
9. ☐ Perform a fixed load voltage drop test on the circuit. See [Fixed-Load Voltage Drop Test](#). Does the circuit pass the test?

a. ☐ **YES** → Install a test SSAM and attempt to clear the fault. If the fault clears, the SSAM may be faulty. For information on replacing the SSAM, see the appropriate procedure:

▪ [Replacement of the Single SAM](#) in the *New Cascadia Workshop Manual*

▪ [Replacement of the Single SAM](#) in the *47X & 49X Workshop Manual*

▪ [Replacement of the ASAM](#) in the *57X Workshop Manual*

b. ☐ **NO** → Repair the circuit, then verify the repair.

NOTE:

Component Code / VMRS = 034-001-088 — WIRING HARNESS - HEADLAMP

Component Code / VMRS = 034-001-135 — HALOGEN CAPSULE - LOW BEAM, HEADLAMP

Component Code / VMRS = 037-001-012 — MODULE, SIGNAL ACTIVATION (SAM), SINGLE

Note:
F07,F15,F16,F18,F25,F26,F35,F36,F41,F42,F43,F44,W12,W13,W14,W15,W16,W17,W18
SPN521080/FMI4
SSAM02T
TROUBLESHOOT
034-001-135,034-001-088,037-001-012

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